



The University
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UC San Diego



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2026
3/6 - 3/10

UNICoRN: Latent Diffusion-based Unified Controllable Image Restoration Network across Multiple Degradations



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Poster
#297

Schedule: Sunday, March 08 | Oral Session 3A (@3pm), Poster Session #2 (@4pm)



SCAN ME

Highlights

The Problem

- Real-world images suffer from *multiple simultaneous degradations* (blur, haze, noise, low-light) — yet no model robustly handles them all unifiedly. Most methods assume prior knowledge of degradation.

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Our Approach: UNICORN

- A controllable latent diffusion model conditioned on **cheap, interpretable low-level visual cues** (transmission maps, edge maps, color maps) via a novel **multi-head ControlNet + Mixture-of-Experts** adapter — no prior knowledge of degradation type required

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Results & Contribution

- Outperforms all prior all-in-one restoration methods by **>10 dB PSNR** on our **new real-world METARESTORE benchmark** — a metalens imaging dataset with compound, entangled degradations
- Surpasses prior methods on synthetic mixed degradation removal, and competitively performs on single degradation restoration

Introduction

Single Degradation Removal



Denoising



Motion Deblurring



Severely low-light condition



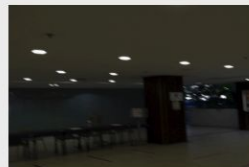
Dehazing



Contrast Enhancement



Deblurring



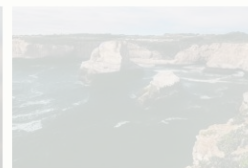
Blur + Haze Removal



Noise + Blur Removal



Blur + *Severe low-light* Enhancement



Real-world Multi-degraded Image Restoration (MetaRestore benchmark)

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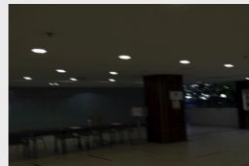
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Synthetic Mixed Degradations



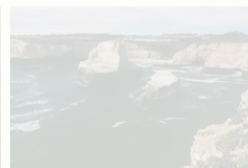
Blur + Haze Removal



Noise + Blur Removal



Blur + *Severe* low-light Enhancement



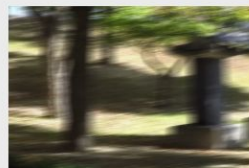
Real-world *Multi-degraded* Image Restoration (MetaRestore benchmark)

Introduction

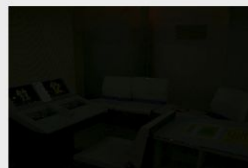
Single Degradation Removal



Denoising



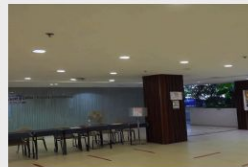
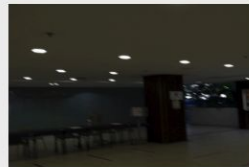
Motion Deblurring



Severely low-light condition



Dehazing

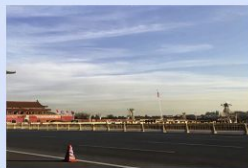


Contrast Enhancement

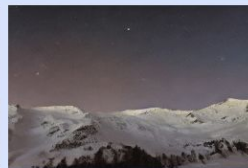


Deblurring

Synthetic Mixed Degradations



Blur + Haze Removal

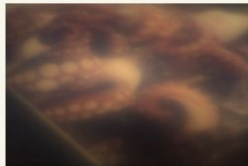


Noise + Blur Removal



Blur + Severe low-light Enhancement

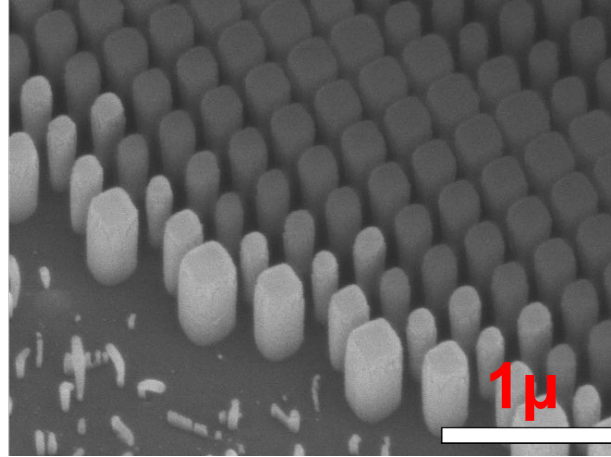
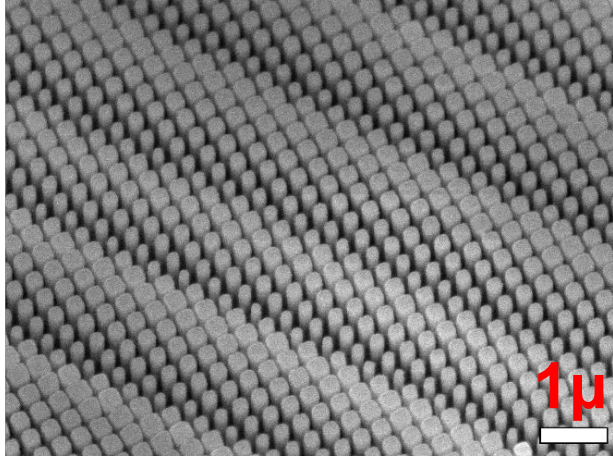
Real-world Complex Degradations



Real-world Multi-degraded Image Restoration (MetaRestore benchmark)

Metalenses

Ultra-thin nano-optical lenses which manipulate light at a subwavelength scale



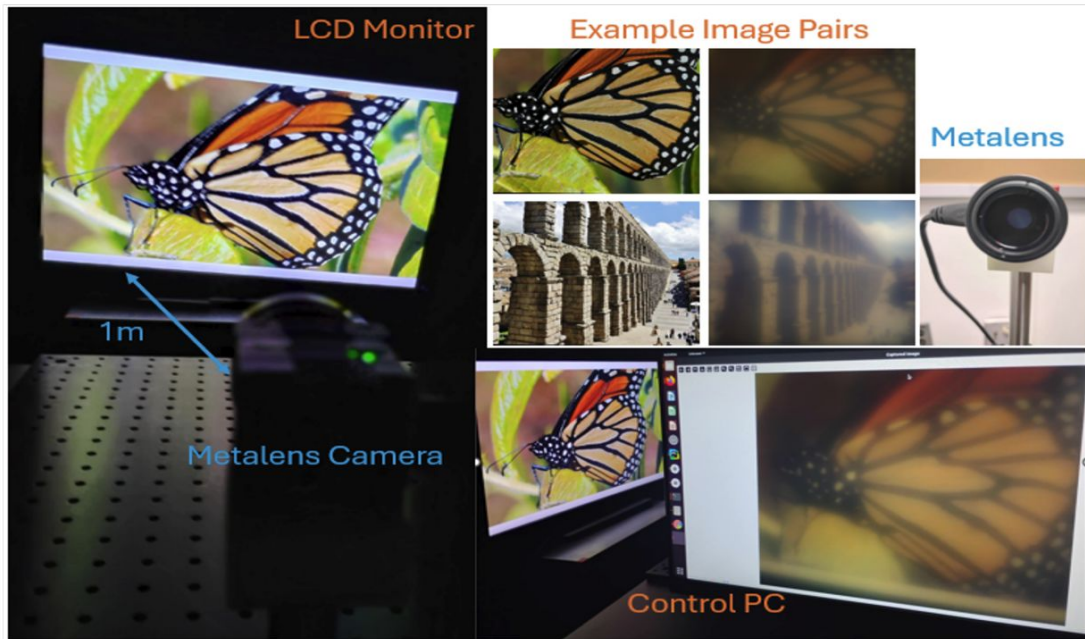
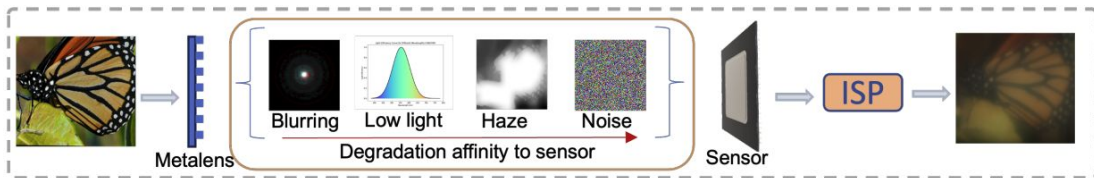
**Regular convex lens
(left) vs metalens (right)**

METARESTORE: Benchmark for Real-World Degradations

Existing datasets for multiple degradation restoration have synthetically blended corruptions.

METARESTORE uses metalens cameras that are prone to simultaneous degradations.

We capture **1200 image pairs** with clean Div2K images + metalens screen captures from an LCD display



Pilot Study: What's wrong with sequential restorations?

Sequential single restoration steps **compound artifacts** and **lose finer details** after each restoration step, irrespective of the order.

Whereas, a unified **one-step restoration** model **faithfully recovers** the HQ image.



Methodology: Outline

① Low-Level Visual Cues

- Extract cheap structural priors directly from the degraded image (*transmission map, edge map, color map, blind restoration estimates*)

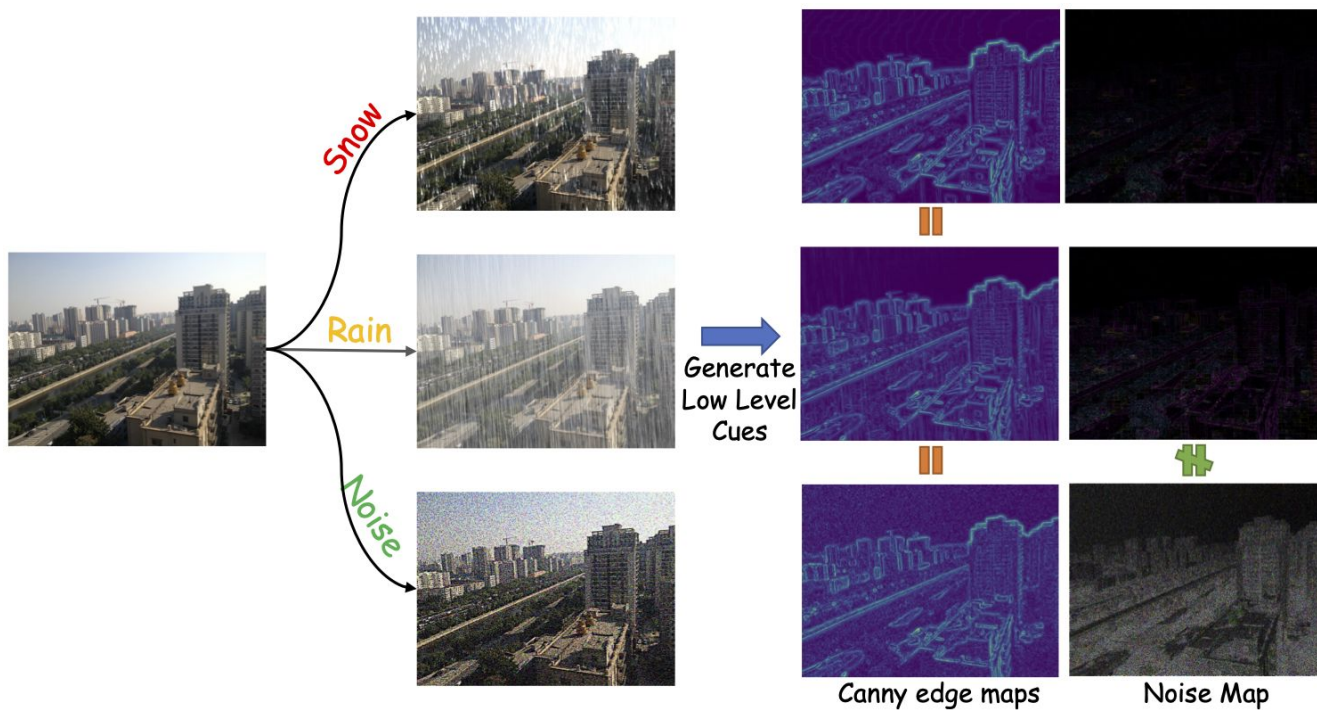
② Multi-Head ControlNet

- One specialized head per degradation → shared frozen Stable Diffusion backbone (de facto standard for visual generation task)

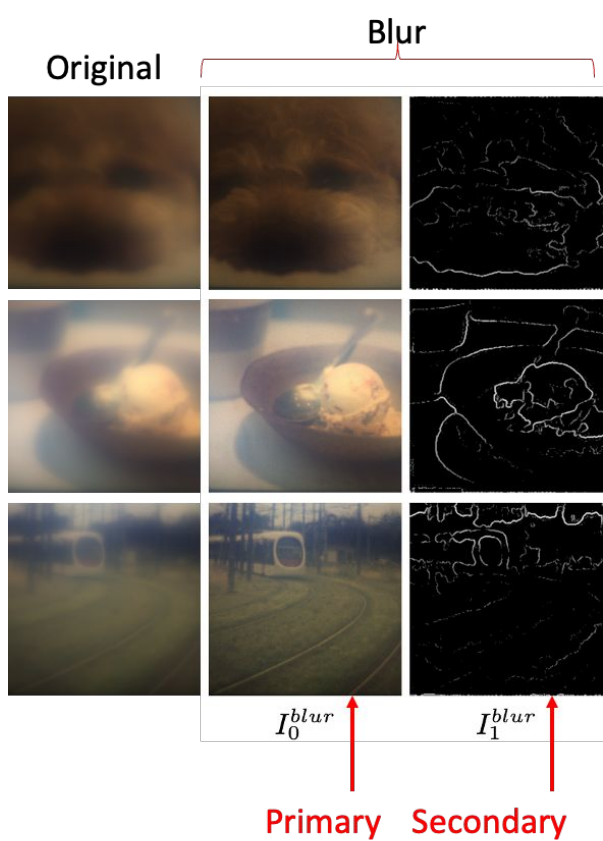
③ Mixture-of-Experts Fusion

- Adaptively weight all heads → single control signal → restored image
- Curriculum learning: single degradations first, then mixed

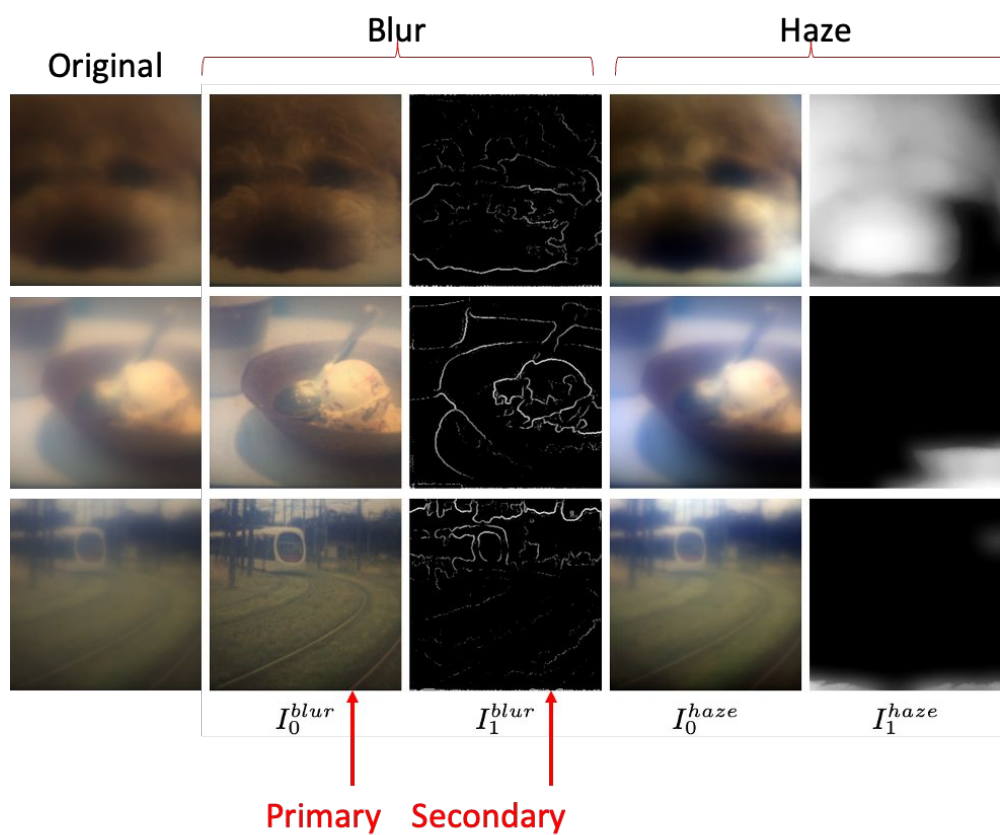
Low-level Visual Cues for Restoration



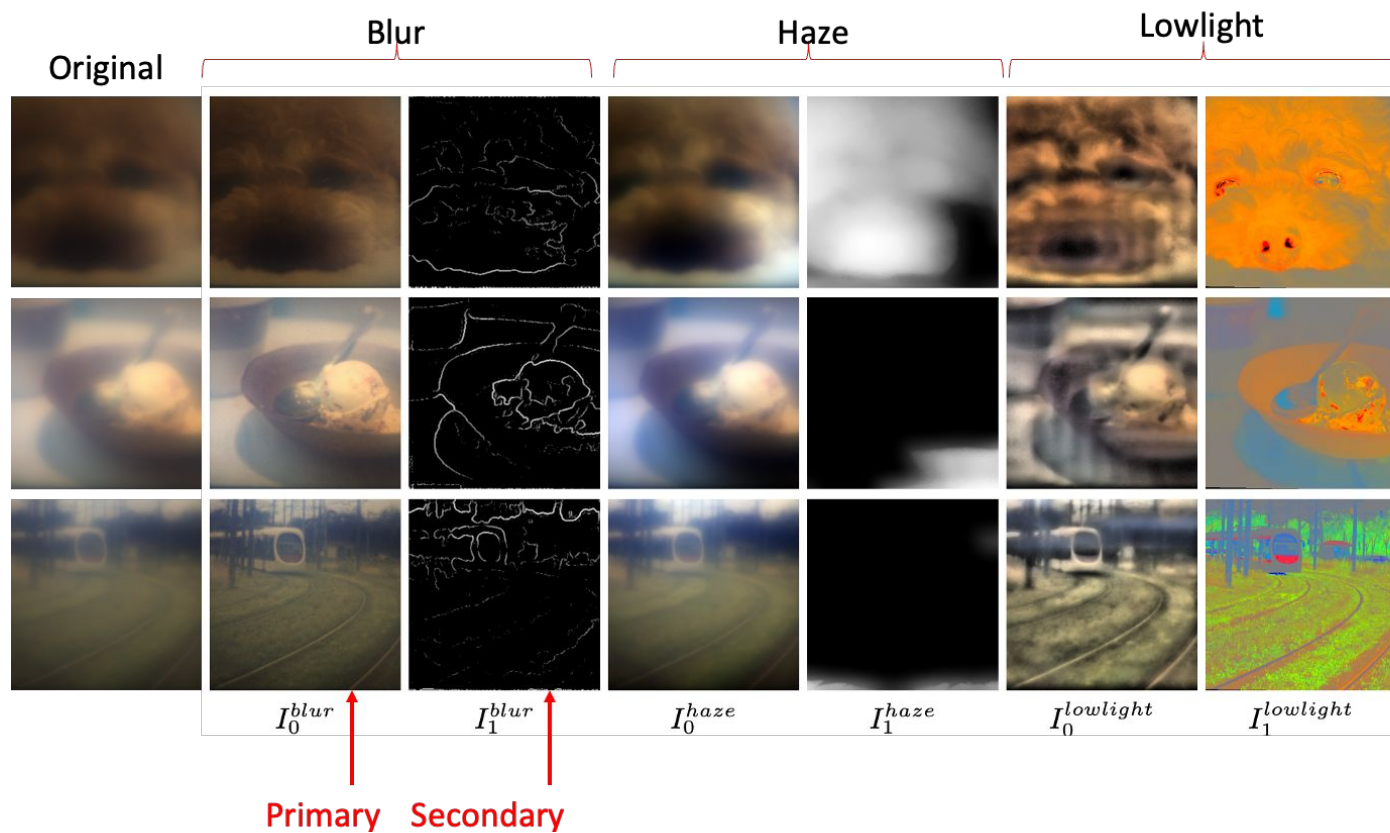
Low-level Visual Cues for Restoration



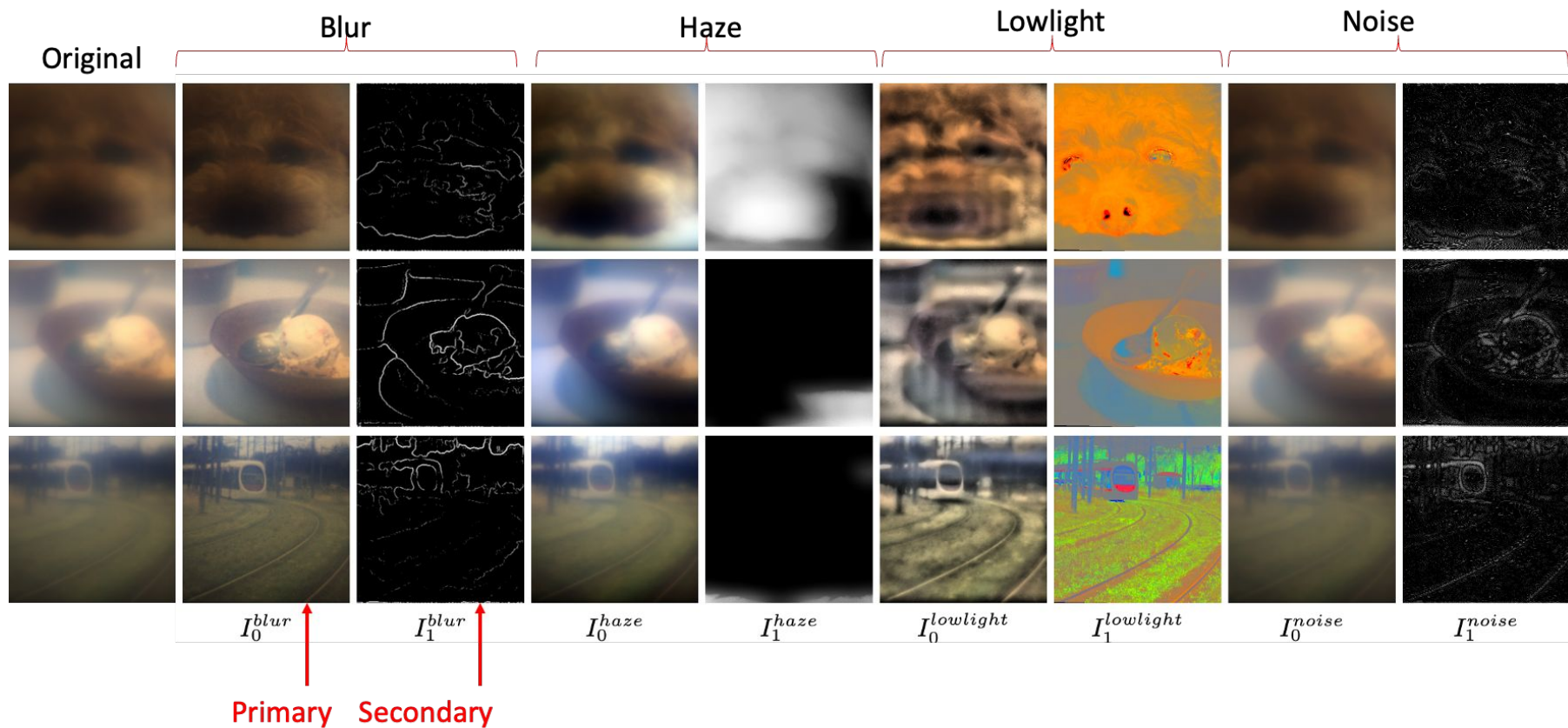
Low-level Visual Cues for Restoration



Low-level Visual Cues for Restoration

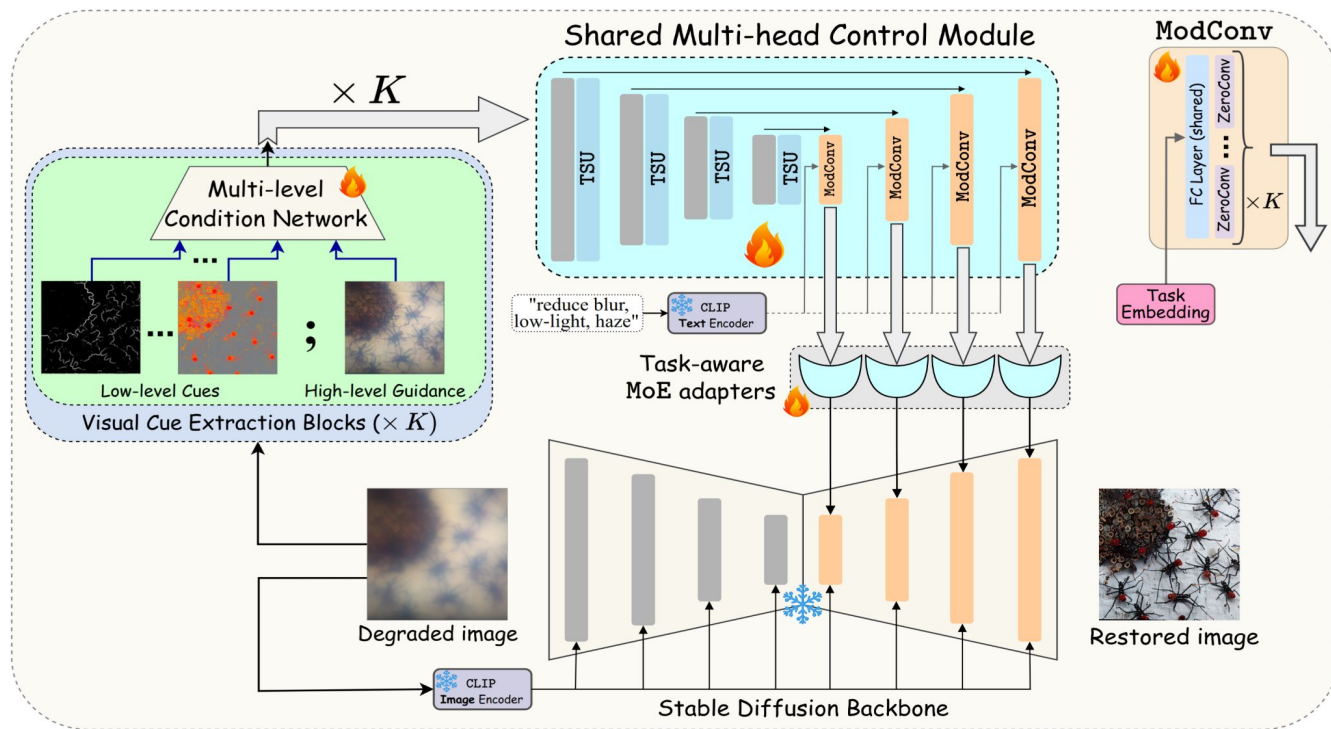


Low-level Visual Cues for Restoration

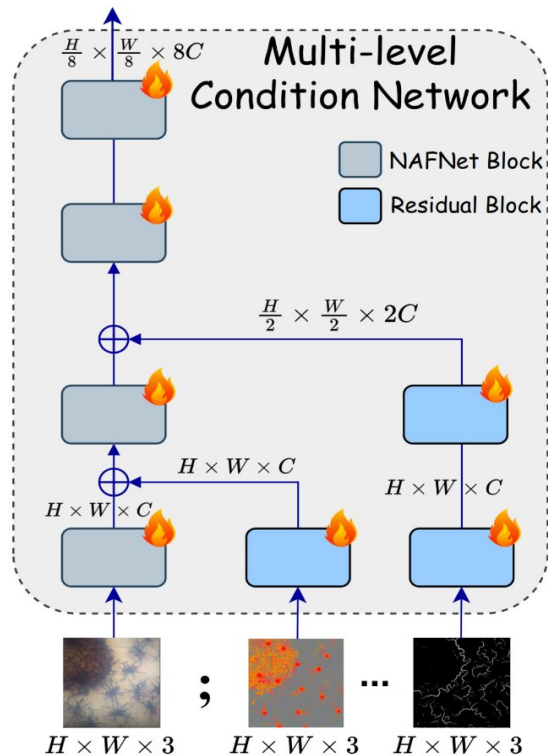


UNICORN: Unified Controllable Restoration Network

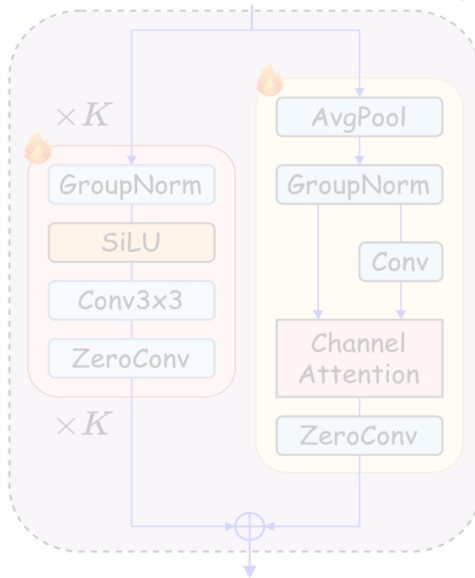
We propose a ControlNet-inspired architecture that leverages low-level visual cues, a shared multi-task control branch, and mixture-of-experts to learn restoration of multiple co-existing degradations in a unified manner.



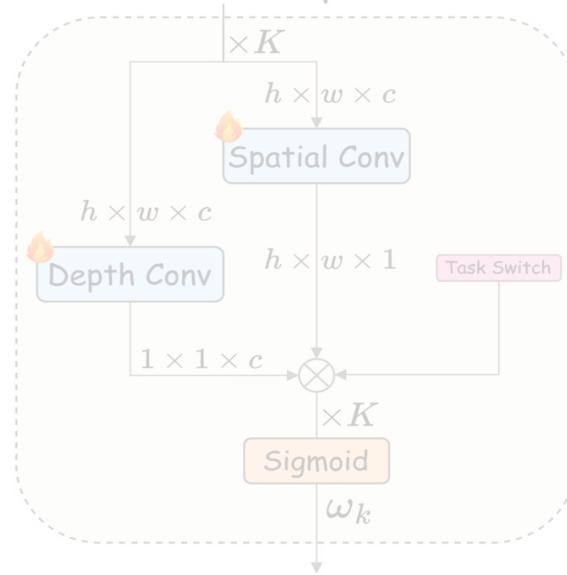
UNICORN: Unified Controllable Restoration Network



Task Stabilizer Unit (TSU)



MoE adapter



Trainable



Frozen

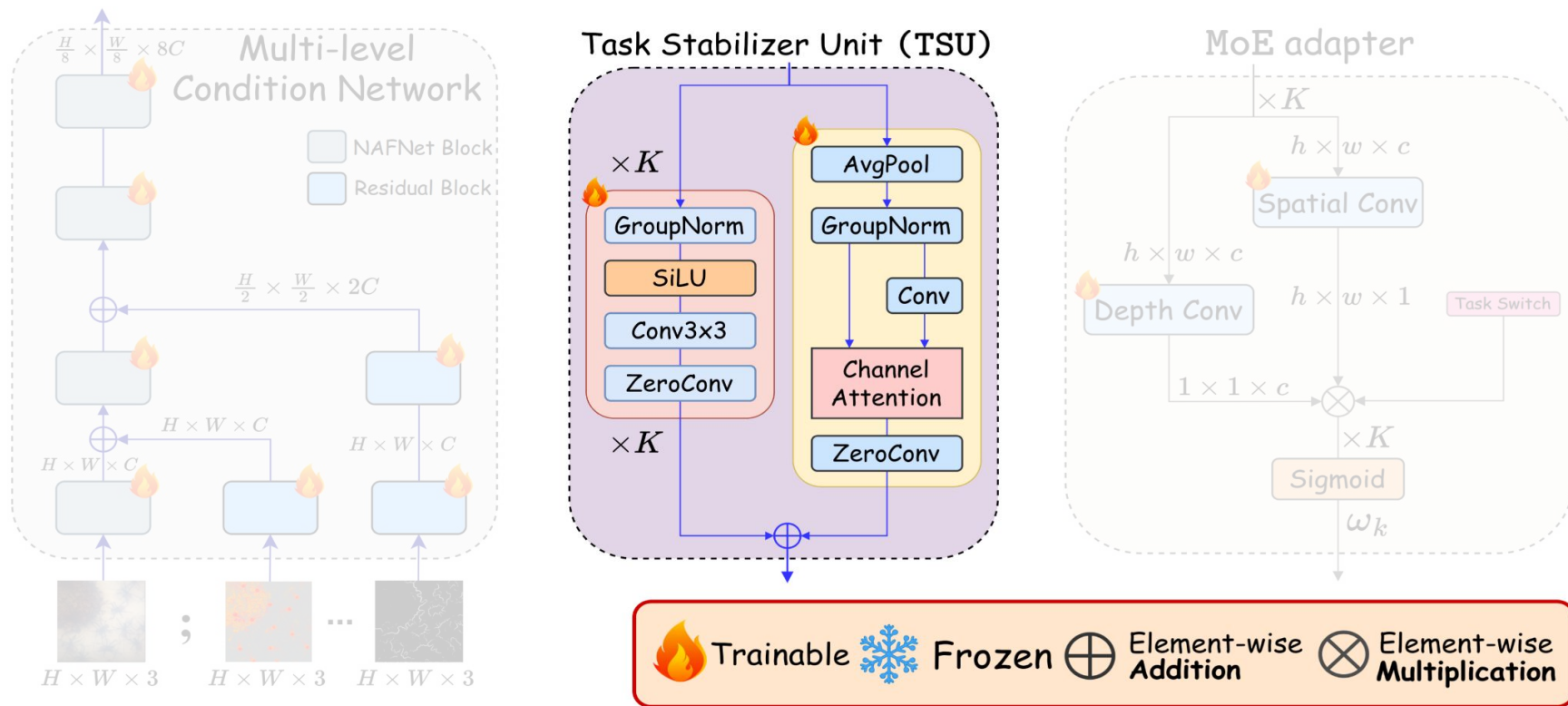


Element-wise
Addition

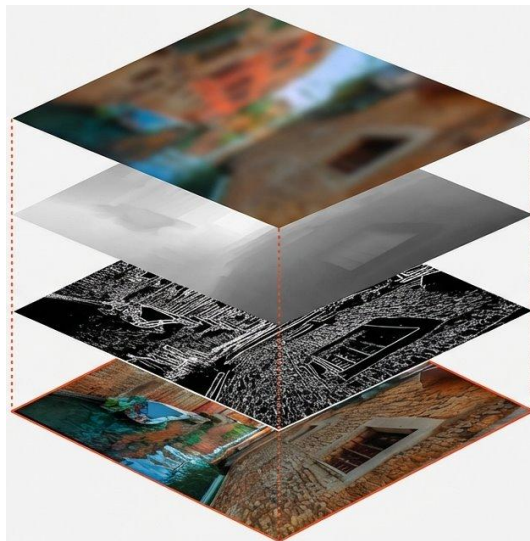


Element-wise
Multiplication

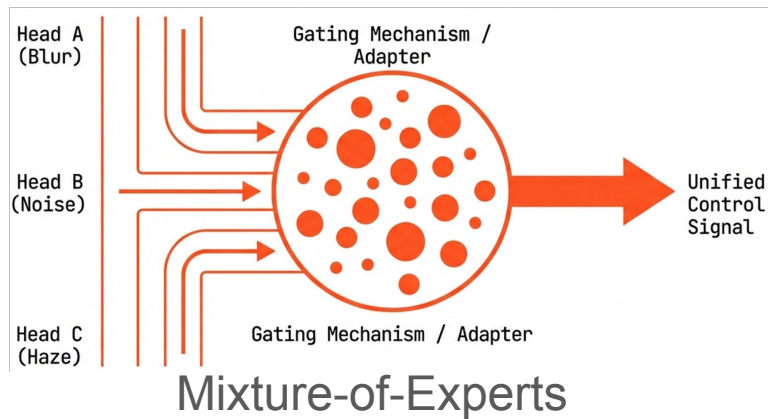
UNICORN: Unified Controllable Restoration Network



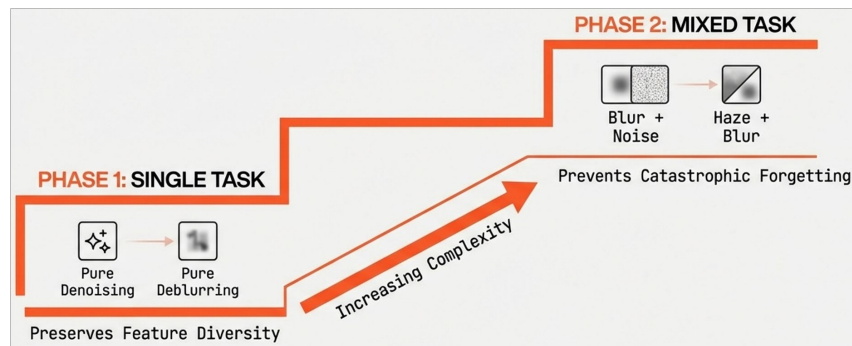
UNICORN: Unified Controllable Restoration Network



Low Level Cues



Mixture-of-Experts

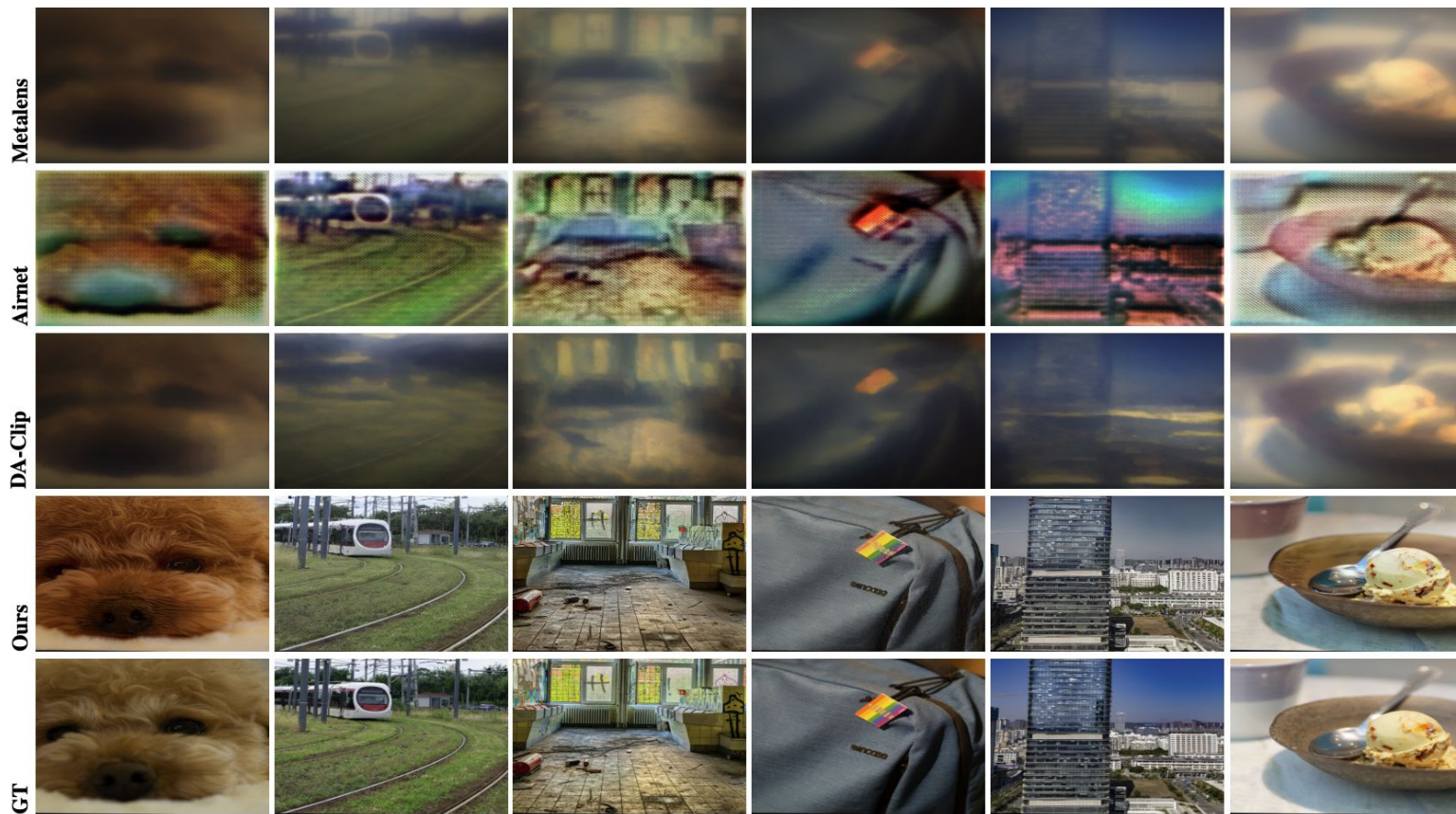


Curriculum Learning

Results: METARESTORE Benchmark

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	NIQE $\downarrow$	BRISQUE $\downarrow$
<i>Zero-shot</i>					
AirNet [43]	12.69	0.341	0.62	29.57	39.96
PromptIR [67]	13.10	0.358	0.616	29.53	34.45
AutoDIR [30]	12.65	0.255	0.654	22.45	61.29
DiffUIR [98]	16.97	0.434	0.472	12.78	47.98
DA-CLIP [52]	14.41	0.312	0.687	22.21	49.16
UNICoRN	27.93	0.436	0.554	5.27	31.66
<i>Fine-tuned</i>					
AirNet [43]	10.88	0.211	0.765	24.29	71.29
MultiWNet [89]	26.05	0.402	0.554	10.19	60.75
UNICoRN	28.54	0.517	0.434	2.85	27.09

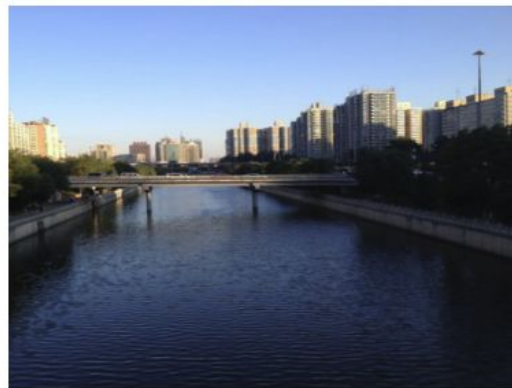
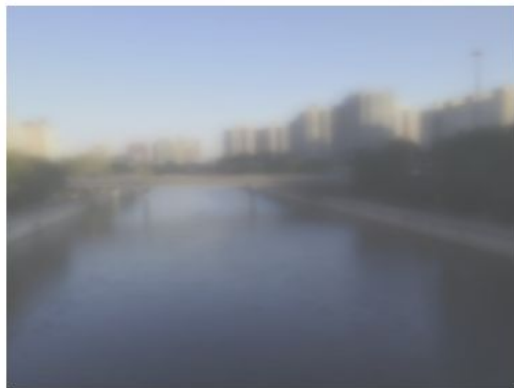
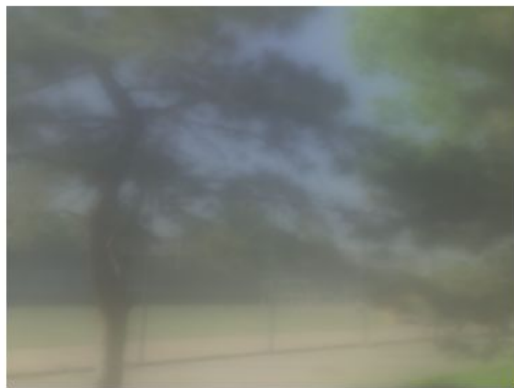
Results: METARESTORE Benchmark



Results: Synthetic Mixed Degradations

Method	<i>Blur + Haze</i> [40]			<i>Noise + Blur: Div2k</i> [1]			<i>Low-light + Blur</i> [99]		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
AutoDIR [30]	12.97	0.394	0.577	17.97	0.475	0.441	18.32	0.662	0.304
AirNet [43]	16.99	0.480	0.569	18.56	0.564	0.401	10.20	0.130	0.515
PromptIR [67]	14.75	0.431	0.528	23.30	0.601	0.455	10.24	0.096	0.501
DA-CLIP [52]	14.37	0.493	0.638	21.52	0.671	0.327	16.44	0.688	0.242
UNICORN	28.83	0.673	0.212	28.55	0.717	0.162	28.47	0.777	0.149

Results: Synthetic Mixed Degradations



Blur-Haze Input

UNICoRN (Ours)

GT

Results: Synthetic Mixed Degradations



Blur-Noise Input

UNICORN (Ours)

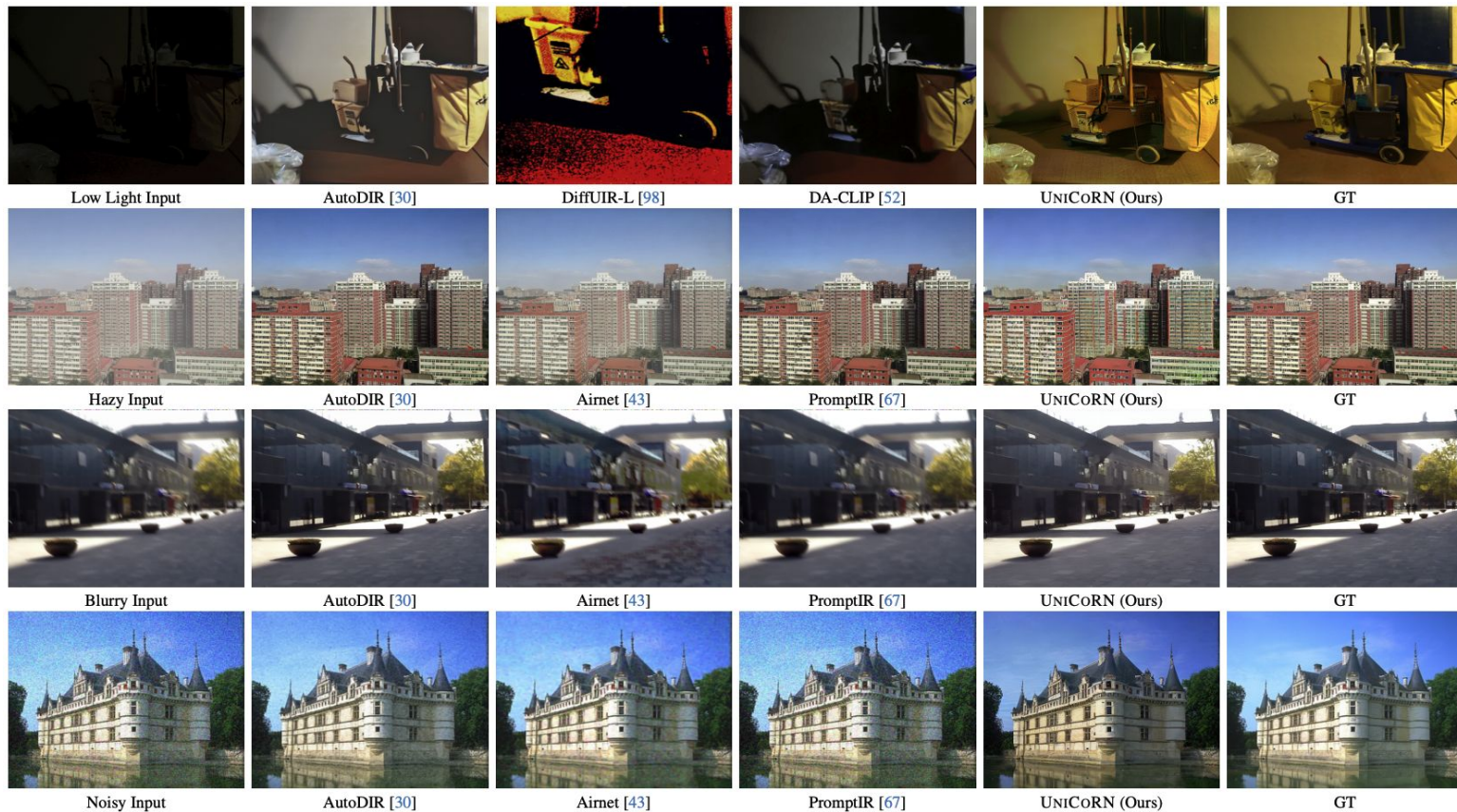
GT

Results: Single Degradation Removal

<i>Dehazing: RESIDE [40]</i>			<i>Motion Deblurring: GoPro [61]</i>		
Method	PSNR $\uparrow$	LPIPS $\downarrow$	Method	PSNR $\uparrow$	LPIPS $\downarrow$
<i>Single-restoration methods</i>					
Restormer [91]	27.09	0.150	MPRNet [56]	31.66	0.091
<i>Non-diffusion approaches</i>					
AirNet [43]	23.70	0.073	AirNet [43]	28.31	0.122
PromptIR [67]	33.11	<u>0.030</u>	PromptIR [67]	31.02	0.131
<i>Diffusion-based approaches</i>					
AutoDIR [30]	20.55	0.025	AutoDIR [30]	28.02	0.167
DiffUIR-L [98]	21.32	0.038	DiffUIR-L [98]	25.12	0.035
DA-CLIP [52]	23.77	0.083	DA-CLIP [52]	27.12	0.134
UniCoRN	29.90	0.174	UniCoRN	28.90	0.155

<i>Low-Light Enhancement: LOL [85]</i>			<i>Denoising: CBSD68 [55]</i>		
Method	PSNR $\uparrow$	LPIPS $\downarrow$	Method	PSNR $\uparrow$	LPIPS $\downarrow$
<i>Single-restoration methods</i>					
LEDNet [99]	25.74	0.224	Restormer [91]	28.59	0.261
<i>Non-diffusion approaches</i>					
AirNet [43]	19.69	0.151	AirNet [43]	26.13	0.394
PromptIR [67]	21.23	0.145	PromptIR [67]	28.82	<u>0.170</u>
<i>Diffusion-based approaches</i>					
AutoDIR [30]	20.98	0.160	AutoDIR [30]	29.24	0.267
DiffUIR-L [98]	25.81	0.040	IR-SDE [51]	28.09	0.103
DA-CLIP [52]	22.09	<u>0.114</u>	DA-CLIP [52]	24.36	0.272
UniCoRN	28.30	0.159	UniCoRN	29.30	0.301

Results: Single Degradation Removal



Ablations on Design Choices

w/o task prompt: model “guesses” which degradations are present and restores them.
Not too significant a drop – shows generalizability.

NAFBlock in Condition Network is a very crucial design choice and benefits our model greatly.

Methods	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
w/o task prompt	27.12	0.670	0.214
w/o secondary guidance	27.45	0.624	0.216
w/o MLCN+NAFBlock	26.55	0.510	0.293
w/o NAFBlock	27.57	0.668	0.242
w/o Curriculum Learning	27.89	0.660	0.190
UNICoRN (<i>full</i>)	28.74	0.772	0.173

Encoder choice	Time to converge (hr)	LPIPS $\downarrow$
Basic encoder	172	0.293
+ ResBlock [44]	144	0.242
+ CABlock [56]	120	0.181
+ NAFBlock [12]	120	0.173

Key Takeaways

- We proposed UNICORN, a latent diffusion model to restore images with multiple coexisting degradations using mixture-of-experts adaptor.
- UNICORN recovers without prior knowledge of degradations present, guided by low-level visual priors (edgemap, colormap, transmission, etc).
- We also introduce METARESTORE, a real-world complex degradation removal benchmark comprising paired clean and severely corrupted images.
- Extensive experiments show our method surpassing several state-of-the-art works on complex, mixed and single degradation removal benchmarks.

Thanks for listening!